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Simulating Differentially Rotating Hybrid Stars

Emilie M. Cote, Pedro L. Espino

Department of Physics and Astronomy, University of California, Berkeley, CA

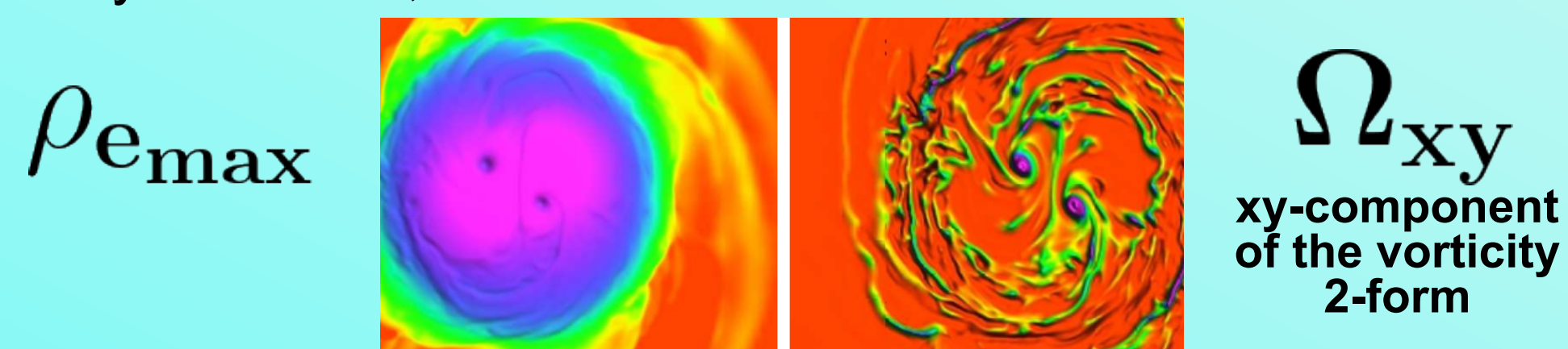


Abstract

Our goal is to understand how deconfinement phase transitions effect the evolution of non-axisymmetric fluid instabilities like the one-arm spiral instability. Studies have shown [3] that when these non-axisymmetric instabilities develop, they can affect the gravitational waves produced by the merger remnants. We are running general relativistic hydrodynamic simulations of differentially rotating stars, that have the potential to be a model for post merger remnants.

One-Arm Spiral Instability

The one-arm Spiral Instability develops when vortices created by the merger cause a toroidal fluid configuration where the max density occurs in a ring around the center-of-mass. The images below show a color map of $\rho_{e\max}$ (equatorial rest-mass density (left) and Ω_{xy} (right) at a select time after the NS collision. When the instability saturates, the $m = 1$ mode dominates.



East, William E., et al. "Relativistic simulations of eccentric binary neutron star mergers: one-arm spiral instability and effects of neutron star spin." [2]

General Relativistic Hydrodynamic Evolution Code Through Numerical Relativity

Our code evolves the stress-energy tensor and the space-time metric by solving Einstein's field equations numerically for small increment in proper time for each block on a discrete grid. Our code works with the following stress-energy tensor (right):

$$\mathbf{T}_{\mu\nu} = \rho_b \mathbf{h} u_\mu u_\nu + \mathbf{P} g_{\mu\nu} \quad \mathbf{G}_{\mu\nu} = 8\pi \mathbf{T}_{\mu\nu}$$

The stress-energy tensor describes the matter and energy content in a system providing the right side of Einstein's equations (left).

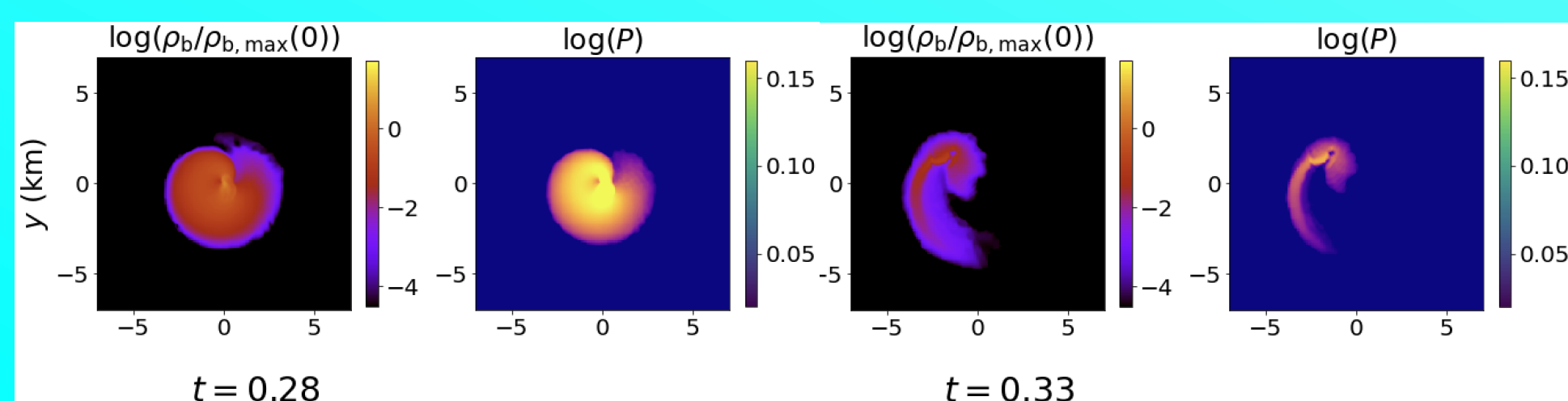
Density Perturbation

Our goal is to excite the one-arm spiral instability explicitly within the simulation. To break angular symmetry, we initiated a density perturbation in the azimuthal direction defined by the added term below.

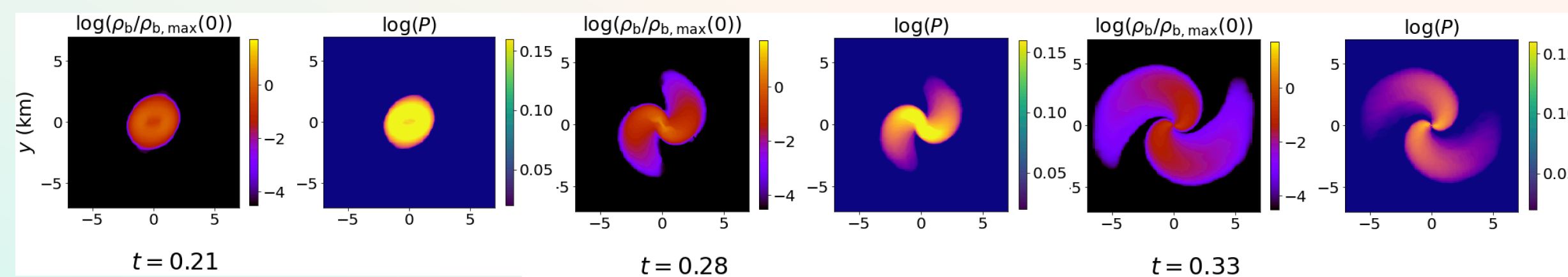
$$\rho_o \longrightarrow \rho_o \left(1 + A \frac{\omega \sin(m\phi)}{r_e} \right)$$

We ran some simulations to test our perturbation for the $m = 1$ and $m = 2$ modes. For these tests, we assumed a $\Gamma = 2$ polytrope.

**Test
for
 $m = 1$
Pert.**



Test for m = 2 Perturbation

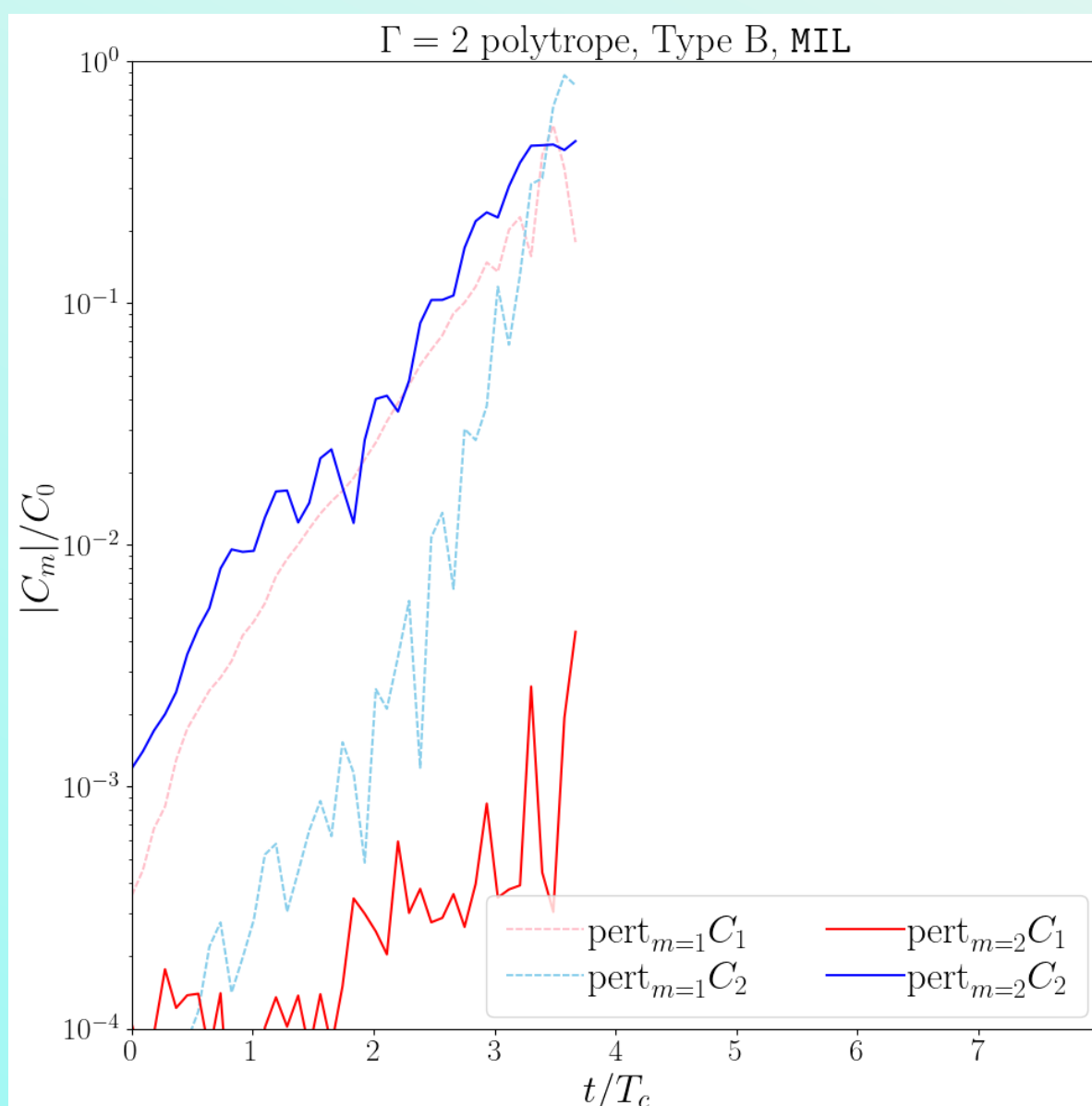
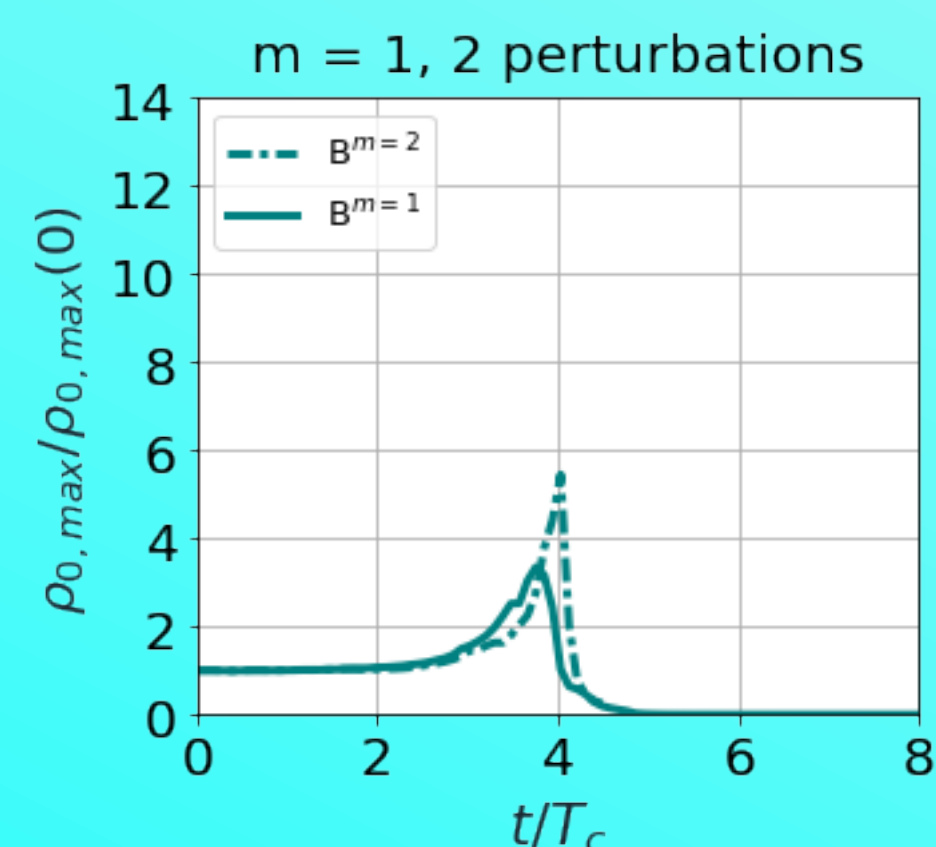
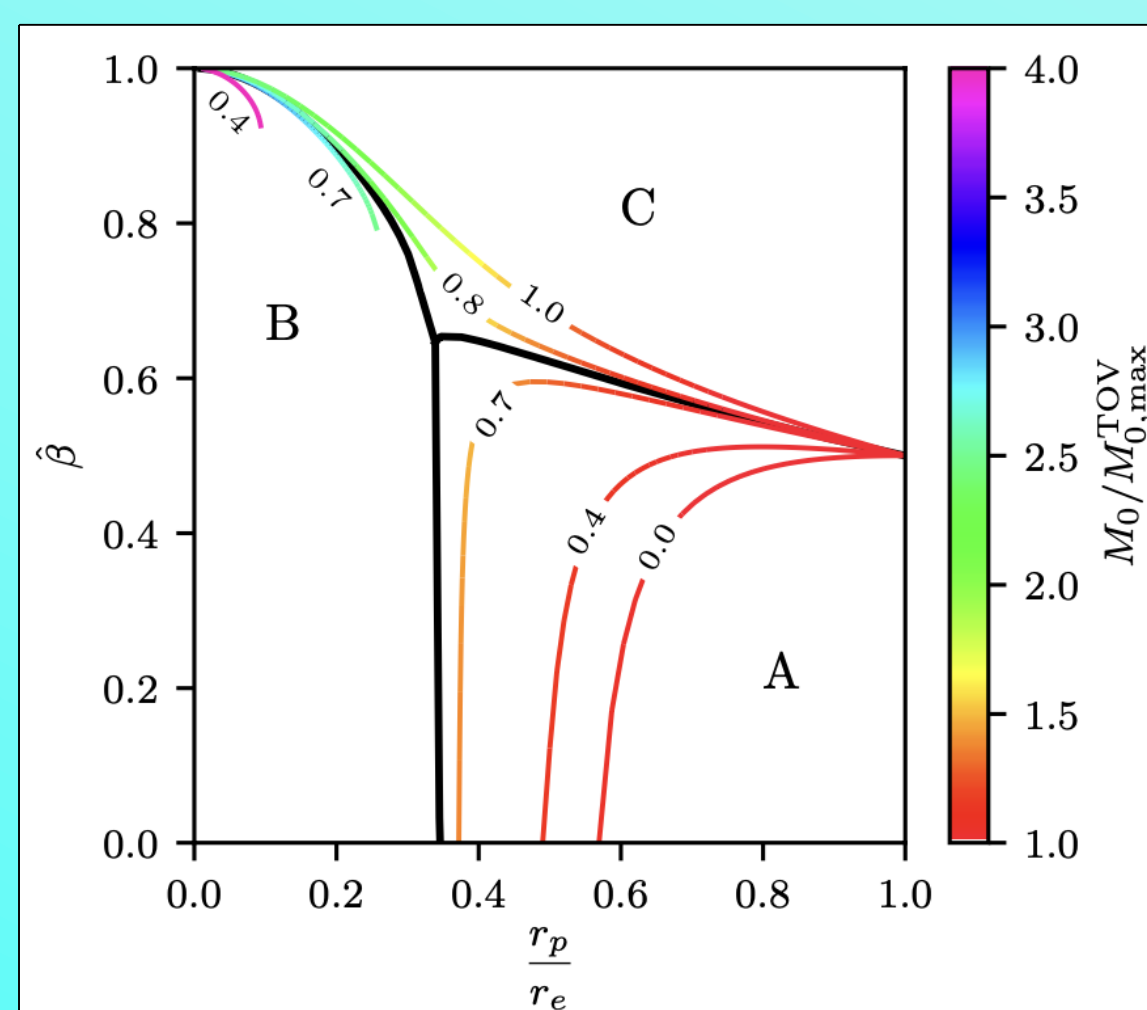


Our code uses finite-temperature EOS tables to create these models. We consider both spheroidal models (max density is at center of star) and quasi-toroidal models (max density occurs in a ring around center of star). In order to get a measure of 'how much' each instability develops, we decompose the density into each mode m and look at their coefficients C_m :

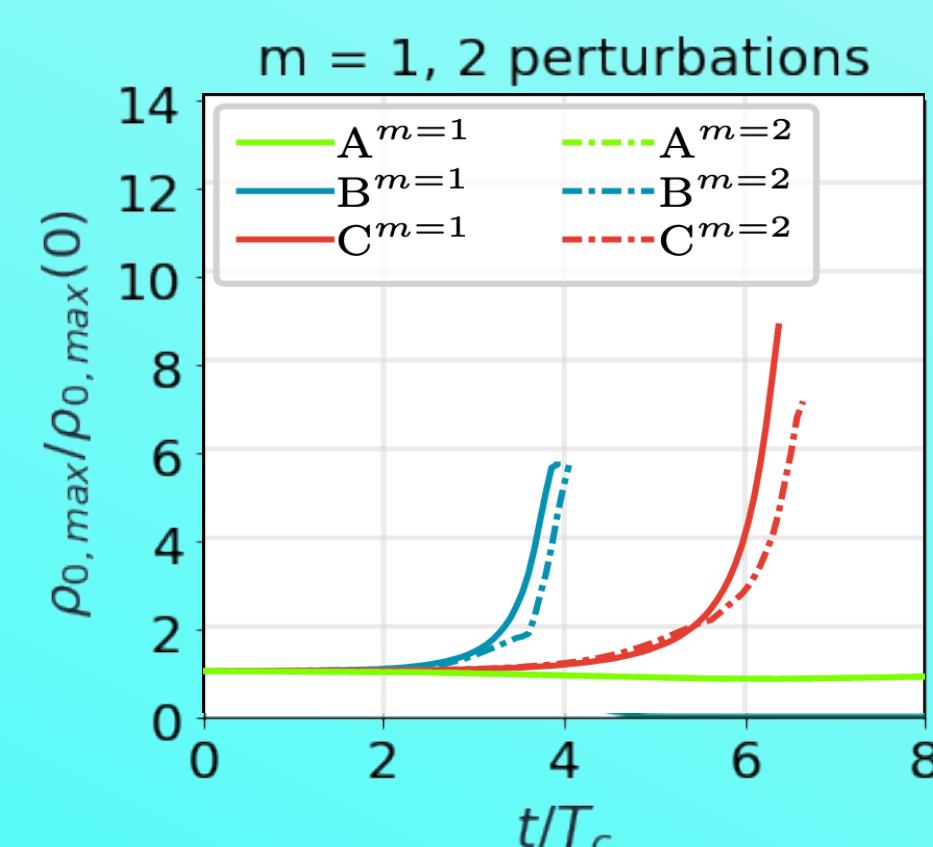
$$\mathbf{C}_m(\varpi, \mathbf{z}) = \frac{1}{2\pi} \int_0^{2\pi} \rho_o \mathbf{u}^o \sqrt{-g} e^{im\phi} d\phi$$

Where, $\varpi = \sqrt{x^2 + y^2}$ is the cylindrical radial and ϕ the azimuthal coord. g is the determinant of the spacetime metric, and u^μ is the fluid 4-velocity. On right are the coefficients

To the right are the coefficients over time of the density decomposition from the $\Gamma = 2$ polytrope perturbation tests previously shown. The solid (dashed) lines correspond to the case where we perturb with an $m=2$ ($m=1$) mode. Since the blue (red) line dominates, we know our simulations worked as intended since this corresponds to the mode we incited. We will now use this perturbation on our full models.

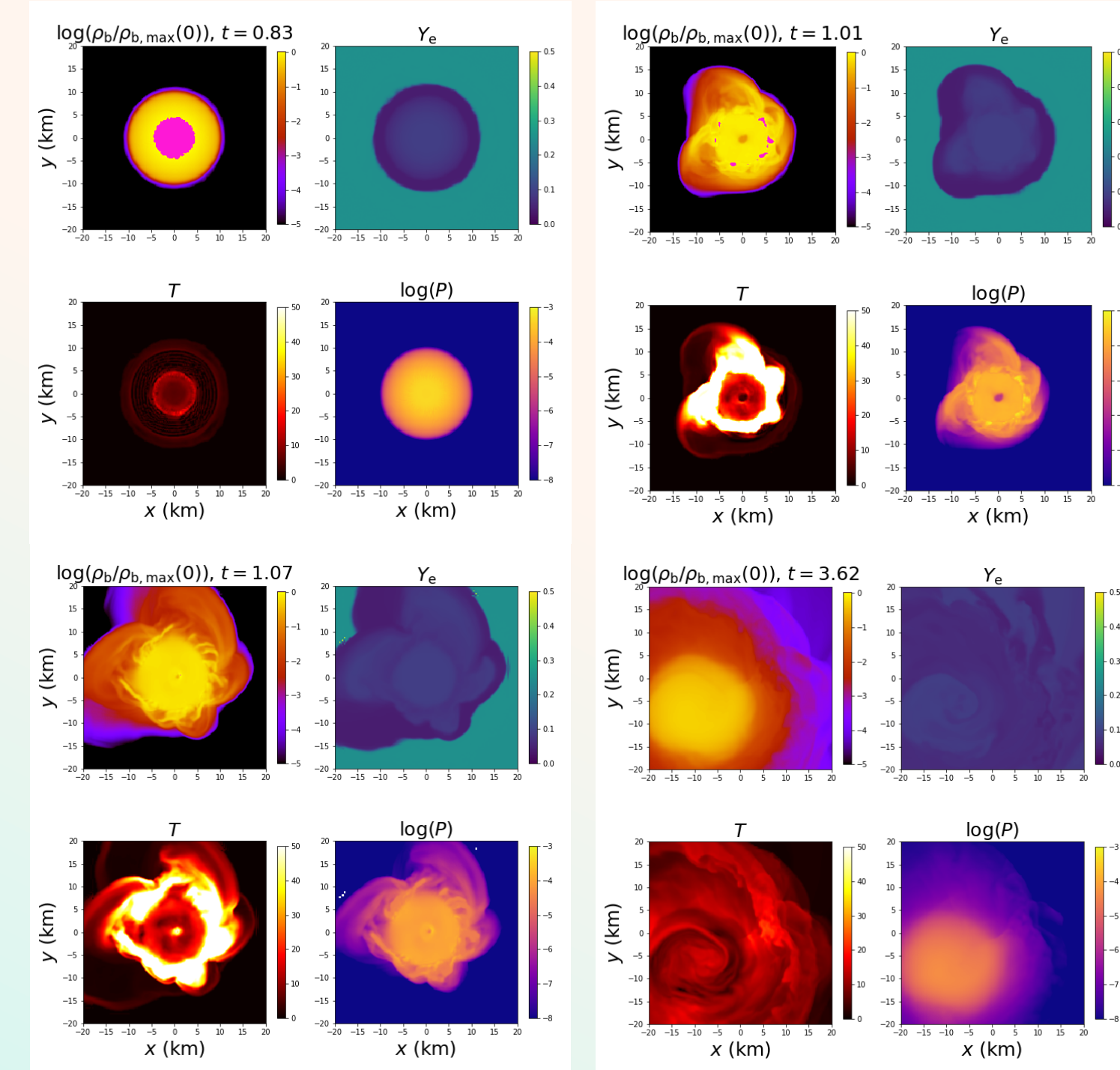


It has been shown [4] that quasi-toroidal NS configuration can appear as transient states after a BNS merger and that these states are unstable to the one-arm spiral instability. We've classified the models into types A, B, and C, whose distinctions are shown in the solution space for a $\Gamma = 2$ polytope projected on the left.



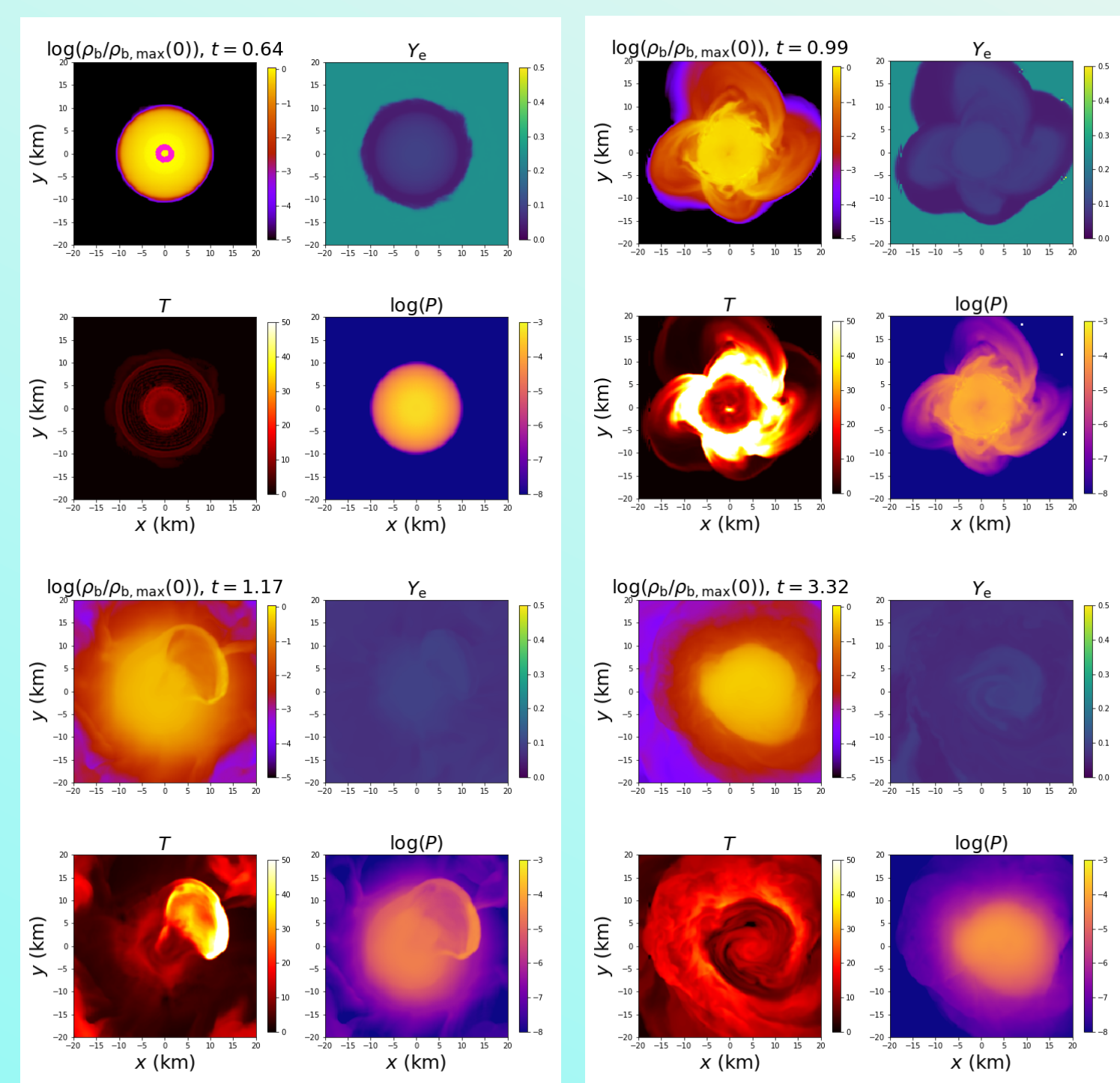
Results

We built a type C model using the **BBKF 1.3 EOS**. We consider two models with the same mass but different radii. Below are snapshots of various fluid variables from our equilibrium simulations (without perturbation to the density for now).



**Model
I**

Both Type C simulations show significant drifts of the center of mass. The development of the one and two-arm spiral instability seemed to be subtly present. In order to determine whether these modes will in-fact develop, **we are now running these simulations with the rest-mass density perturbation.**



**Model
II**

References

- [2] W. E. East, V. Paschalidis, F. Pretorius, and S. L. Shapiro, "Relativistic simulations of eccentric binary neutron star mergers: One-arm spiral instability and effects of neutron star spin," Physical Review D, vol. 93, jan 2016.
- [3] P. L. Espino, A. Prakash, D. Radice, and D. Logoteta, "Revealing phase transition in dense matter with gravitational wave spectroscopy of binary neutron star mergers," 2023.
- [4] P. L. Espino, V. Paschalidis, T. W. Baumgarte, and S. L. Shapiro, "Dynamical stability of quasitoroidal differentially rotating neutron stars," Physical Review D, vol. 100, aug 2019.